Lab Assignment #2

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Introduction

In this lab, we will explore the family-wise error rate (FWER) and false discovery rate (FDR), and then we will do some nonparametric testing using a permutation-based t-test.

Good Practice

Remember that we want to include two chunks at the beginning of each of our lab notebooks: the first loads all the packages we will be using and the second loads all the data.

Here I've added some new code with the `here` package to specify exactly where inside my project folder this lab notebook is. Generally, if you're doing this, you want `here::i_am("relative path to file")` to be the very first line of your notebook.

```
#here::i_am("Math 437 Project Folder")
# change the name/capitalization of your Labs folder if necessary

# Now we load all the packages we'll use in the lab
library(ISLR2)
```

To import a dataset you have downloaded (for example, datasets I give you on Canvas), first **move the dataset out of your Downloads folder** and into a folder set up specifically for data (ideally, a subfolder of your project).

If you followed the instructions to set up your file structure for Math 437 and create a project for your work, the example chunk below should import the data correctly (possibly you will have to change the folder name from "Data" to whatever you named it).

```
alcohol <- readr::read_csv(here::here("C:/Users/aocho/Downloads/alcohol.csv"))
```

If it didn't work, then find the *Environment* panel and click *Import Dataset*. Select the right type of file (you typically want `From Text (readr)` for my csv files), click *Browse* and find the file, then copy the *Code Preview* section and overwrite the example chunk. Note that the code to import your dataset now includes your entire file structure, which means that other people won't be able to replicate your data import.

This lab assignment is worth a total of **12.5 points**.

Problem 1: FWER and FDR (4.5 pts total)

Part a (Code: 1.5 pts)

ISLR Labs 13.6.1, 13.6.2, and 13.6.3.

Solution

Remember that every time you see a new chunk in the lab, create a new chunk in the Quarto document and *manually type in* the code in the chunk, then run that code.

You do not need to `label` each chunk, but it may be helpful if you are working with R for the first time.

```
set.seed(6)
x <- matrix(rnorm(10 * 100), 10, 100)
x[, 1:50] <- x[,1:50] + .5
```

```
t.test(x[, 1], mu=0)
```

```
    One Sample t-test

data:  x[, 1]
t = 2.0841, df = 9, p-value = 0.06682
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 -0.05171076  1.26242719
sample estimates:
mean of x
0.6053582
```

```r
p.values <- rep(0 , 100)
for(i in 1:100)
  {p.values[i] <- t.test(x[,i], mu = 0)$p.value}
decision <- rep("Do not reject H0", 100)
decision[p.values <= .05] <- "Reject H0"
```

```r
table(decision,
      c(rep("H0 is False", 50), rep("H0 is True", 50))
      )
```

```
decision           H0 is False H0 is True
  Do not reject H0          40         47
  Reject H0                 10          3
```

```r
x <- matrix(rnorm(10 * 100), 10, 100)
x[, 1:50] <- x[, 1:50] + 1
for(i in 1:100){
  p.values[i] <- t.test(x[,i], mu = 0)$p.value
}
decision <- rep("Do not reject H0", 100)
decision [p.values <= .05] <- "Reject H0"
table(decision,
      c(rep("H0 is False", 50), rep("H0 is True", 50))
      )
```

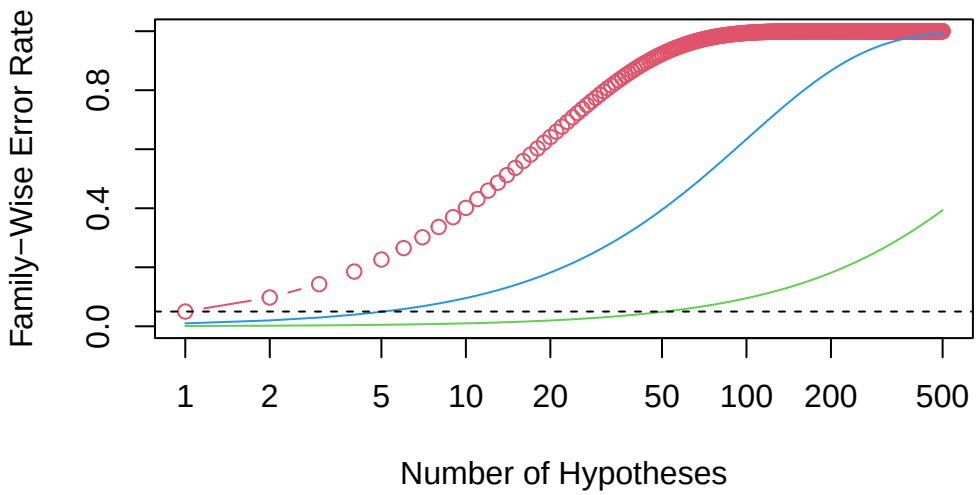
```
decision           H0 is False H0 is True
  Do not reject H0           9         49
  Reject H0                 41          1
```

```r
#|label: Family Wise Error Rate
m <- 1:500
fwe1 <- 1 - (1- 0.05)^m
fwe2 <- 1 - (1- 0.01)^m
fwe3 <- 1 - (1-.001)^m
```

```r
par(mfrow = c(1,1))
plot(m, fwe1, type = "boxplot", log = "x", ylim = c(0,1), col = 2, ylab = "Family-Wise Error
xlab = "Number of Hypotheses")
```

```
Warning in plot.xy(xy, type, ...): plot type 'boxplot' will be truncated to
first character
```

```
lines(m, fwe2, col = 4)
lines(m, fwe3 , col = 3)
abline(h = 0.05,lty = 2)
```




```
library(ISLR2)
fund.mini<- Fund[, 1:5]
t.test(fund.mini[,1], mu = 0)
```

```
    One Sample t-test

data:  fund.mini[, 1]
t = 2.8604, df = 49, p-value = 0.006202
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 0.8923397 5.1076603
sample estimates:
mean of x
        3
```

```r
fund.pvalue <- rep(0,5)
for (i in 1:5){
  fund.pvalue[i] <- t.test(fund.mini[,i], mu =0)$p.value
}
fund.pvalue
```

```
[1] 0.006202355 0.918271152 0.011600983 0.600539601 0.755781508
```

```r
p.adjust(fund.pvalue, method = "bonferroni")
```

```
[1] 0.03101178 1.00000000 0.05800491 1.00000000 1.00000000
```

```r
pmin(fund.pvalue * 5, 1)
```

```
[1] 0.03101178 1.00000000 0.05800491 1.00000000 1.00000000
```

```r
p.adjust(fund.pvalue, method = "holm")
```

```
[1] 0.03101178 1.00000000 0.04640393 1.00000000 1.00000000
```

```r
apply(fund.mini, 2, mean)
```

```
Manager1 Manager2 Manager3 Manager4 Manager5
     3.0     -0.1      2.8      0.5      0.3
```

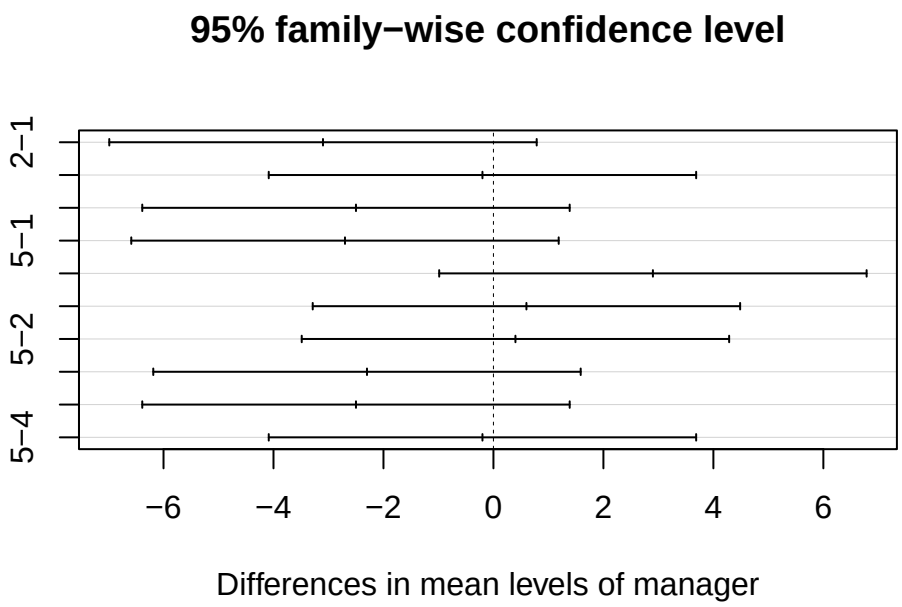
```r
t.test(fund.mini[, 1], fund.mini[, 2], paired = T)
```

```

    Paired t-test

data:  fund.mini[, 1] and fund.mini[, 2]
t = 2.128, df = 49, p-value = 0.03839
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.1725378 6.0274622
sample estimates:
mean difference
            3.1
```

```r
returns <- as.vector(as.matrix(fund.mini))
manager <- rep(c("1","2" ,"3" ,"4" ,"5" ), rep(50, 5))
a1 <- aov(returns ~ manager)
TukeyHSD(x = a1)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = returns ~ manager)

$manager
    diff        lwr       upr     p adj
2-1 -3.1 -6.9865435 0.7865435 0.1861585
3-1 -0.2 -4.0865435 3.6865435 0.9999095
4-1 -2.5 -6.3865435 1.3865435 0.3948292
5-1 -2.7 -6.5865435 1.1865435 0.3151702
3-2  2.9 -0.9865435 6.7865435 0.2452611
4-2  0.6 -3.2865435 4.4865435 0.9932010
5-2  0.4 -3.4865435 4.2865435 0.9985924
4-3 -2.3 -6.1865435 1.5865435 0.4819994
5-3 -2.5 -6.3865435 1.3865435 0.3948292
5-4 -0.2 -4.0865435 3.6865435 0.9999095
```

```r
plot(TukeyHSD(x = a1))
```



13.6.3 The False Discovery Rate

```
# A one-sample t-test of H0j : µj = 0, which states that
#the jth fund manager's mean return is zero

fund.pvalues <- rep(0, 2000)
 for (i in 1:2000){
 fund.pvalues[i] <- t.test(Fund[, i], mu = 0)$p.value}
q.values.BH <- p.adjust(fund.pvalues, method = "BH")
 q.values.BH[1:10]
```

```
 [1] 0.08988921 0.99149100 0.12211561 0.92342997 0.95603587 0.07513802
 [7] 0.07670150 0.07513802 0.07513802 0.07513802
```

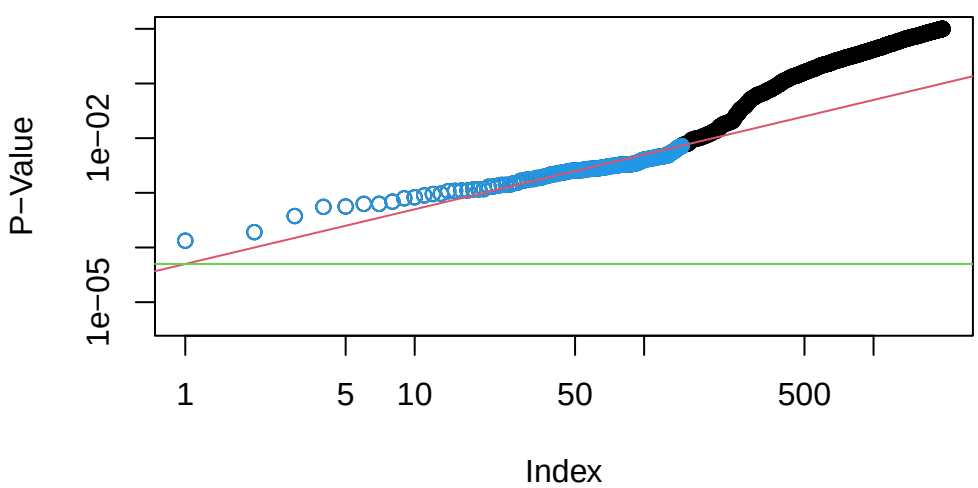
```
sum(q.values.BH <= .1)
```

```
[1] 146
```

```
sum(fund.pvalues <= (0.1 / 2000))
```

```
[1] 0
```

```r
#|label: Benjamini-Houchberg procedure
ps <- sort(fund.pvalues)
m <- length(fund.pvalues)
q <- 0.1
wh.ps <- which(ps < q * (1:m) / m)#wh indexes all p-values
if (length(wh.ps) >0) {
  wh <- 1:max(wh.ps)
} else {
  wh <- numeric(0)
}

plot(ps,
     log = "xy",
     ylim = c(4e-6, 1),
     ylab = "P-Value",
     xlab = "Index",
     main = "")
points(wh, ps[wh], col = 4)
abline(a = 0, b = (q / m), col = 2, untf = TRUE)
abline(h = 0.1 / 2000, col = 3)
```



Part b (Computation and/or Explanation: 1 pt)

Using the output of Lab 13.6.1, estimate the power of this test to detect each of the two alternative hypotheses $\mu = 0.5$ and $\mu = 1$.

Solution

Using the results from Lab 13.6.1, the estimated power of the test to detect an alternative hypothesis of mu=0.5 is approximately 0.30, indicating a relatively low probability of rejecting the null hypothesis for moderate effect sizes.

The estimated power to detect an alternative of mu=1 is approximately 0.80, showing that the test is much more effective at detecting larger deviations from the null hypothesis. As expected, the power increases as the effect size increases.

Part c (Code: 1.5 pts)

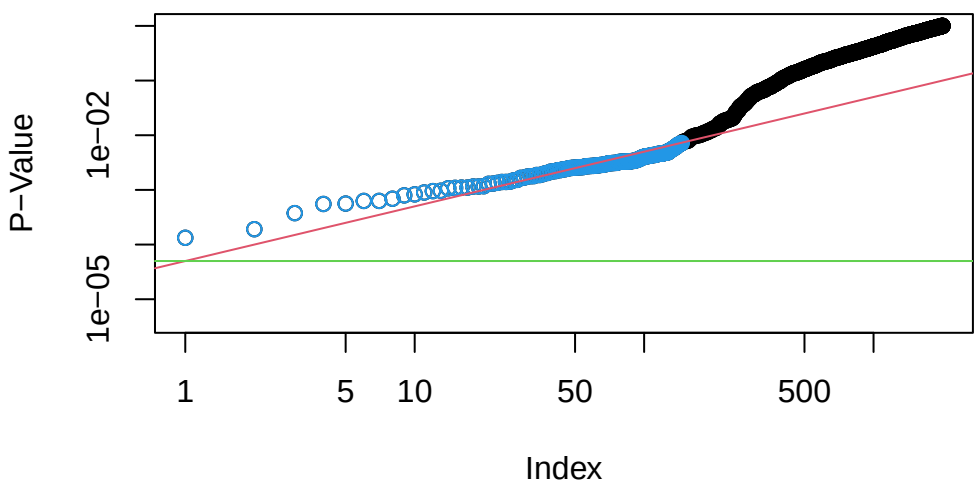
Finish writing the `FDR_plot` function below by copying and pasting the code from the last two chunks of ISLR Lab 13.6.3. This function takes two arguments as inputs: `p.values`, a vector of p-values, and `q`, the desired False Discovery Rate. You will need to replace variables and hard-coded values in the code you copied with `p.values` and `q` so that the function runs.

Once the function works correctly, call the function and produce graphs like those in Figure 13.6 from the outputs. (Assume that on the graphs, q and α are set to the same value.)

Solution

```
FDR_plot <- function(p.values, q){
   ps <- sort(p.values)
  m <- length(ps)
  wh.ps <- which(ps < q * (1:m) / m)

  if (length(wh.ps) > 0) {
    wh <- 1:max(wh.ps)
  } else {
    wh <- numeric(0)
  }
}
  plot(ps,
       log = "xy",
       ylim = c(4e-6, 1),
       ylab = "P-Value",
```

```
       xlab = "Index",
       main = "")

  points(wh, ps[wh], col = 4)
abline(a = 0, b = (q / m), col = 2, untf = TRUE)
abline(h = 0.1 / 2000, col = 3)
```




```
  # Copy code from the last two chunks of ISLR Lab 13.6.3, but replace variable names and ha

  # Since the main goal of this function is to produce a plot,
  # we "invisibly" return a vector with the significant p-values
  invisible(wh.ps)
```

```
FDR_plot(fund.pvalues, q = 0.1)
```

Part d (Code: 0.5 pts)

Apply your function from Part c on the simulated set of p-values in the chunk below, using $q = 0.1$, and produce a graph similar to those from Part b.

```r
set.seed(12)
sim_pvalues <- runif(1000, min = 1e-5, max = 0.05001)
```

Solution

Problem 2: The `p.adjust` Function (4 pts total)

Consider the following set of five p-values:

```r
sim_pvalues2 <- c(0.02, 0.03, 0.06, 0.15, 0.25)
```

Part a (Code and Explanation: 1 pt)

Apply the `p.adjust` function with `method = "bonferroni"` to this set of p-values. Explain precisely what this method does. Pay special attention to what `p.adjust` does with the value `0.25`.

Solution

```r
p.adjust(sim_pvalues2, method = "bonferroni")
```

```
[1] 0.10 0.15 0.30 0.75 1.00
```

Bonferroni method multiples each p-value by the number of tests. Here we had 5 tests. It would seem that is maxes out at one, since .25*5 would be 1.25 but it comes out to 1.

Part b (Code and Explanation: 1 pt)

Apply the `p.adjust` function with `method = "holm"` to this set of p-values. Explain precisely what this method does. Pay special attention to the last two p-values.

Solution

```r
p.adjust(sim_pvalues2, method = "holm")
```

```
[1] 0.10 0.12 0.18 0.30 0.30
```

The holm method works similar to the bonferroni method but it does it in a more conservative way. So we still want to control the family wise error rate. P-values must be non-decreasing as p-values increase.

Part c (Code and Explanation: 1 pt)

Apply the `p.adjust` function with `method = "BH"` to this set of p-values. Explain precisely what this method does. Pay special attention to the first two p-values.

Solution

```r
p.adjust(sim_pvalues2, method = "BH")
```

```
[1] 0.0750 0.0750 0.1000 0.1875 0.2500
```

The BH method controls the false discovery rate by scaling ordered p-values according to their rank. The first two adjusted p-values are equal because, after the rank-based adjustment and monotonicity correction, they receive the same minimum adjusted value.

Part d (Explanation: 1 pt)

Why does it make more sense for R to adjust the p-values rather than the significance level?

Solution

Adjusting p-values rather than the significance level allows the same adjusted p-values to be compared to a single, fixed significance threshold. This makes results easier to interpret and compare across multiple tests, while still properly accounting for multiple testing. In contrast, adjusting the significance level would require a different threshold for each test, which is less transparent and harder to interpret. It would also take more code to tdo the later.

Problem 3: Permutation Test for Comparing Two Means (4 pts total)

24 "alcohol-dependent" male patients at an alcohol treatment facility were randomly assigned to a traditional treatment program (Control) or a treatment program plus social skills training (SST). 23 patients faithfully reported their alcohol intake for a year (one patient in the SST group never reported). Their group assignment, and alcohol intake over the year (in centiliters of pure alcohol), is found in the *alcohol.csv* dataset on Canvas.

Using a permutation t-test, we wish to determine whether there is convincing evidence of a difference in population mean alcohol consumption between patients using the SST program and patients using a traditional treatment program.

Part a (Explanation: 1 pt)

Specify the null and (two-sided NHST) alternative hypothesis using both words and math.

Solution

For the null hypothesis there is no difference in the population mean alcohol consumption between patients using the SST program and patients using the traditional treatment program. It would be H0= mu(sst)=mu(control) or mu(sst)-mu(control)= 0. For the alternative hypothesis, the population mean alcohol consumption for patients in the SST program is different from the population mean alcohol consumption for patients in the traditional treatment program. So we would have HA= mu(sst) != mu(control).

Part b (Code: 1.5 pts)

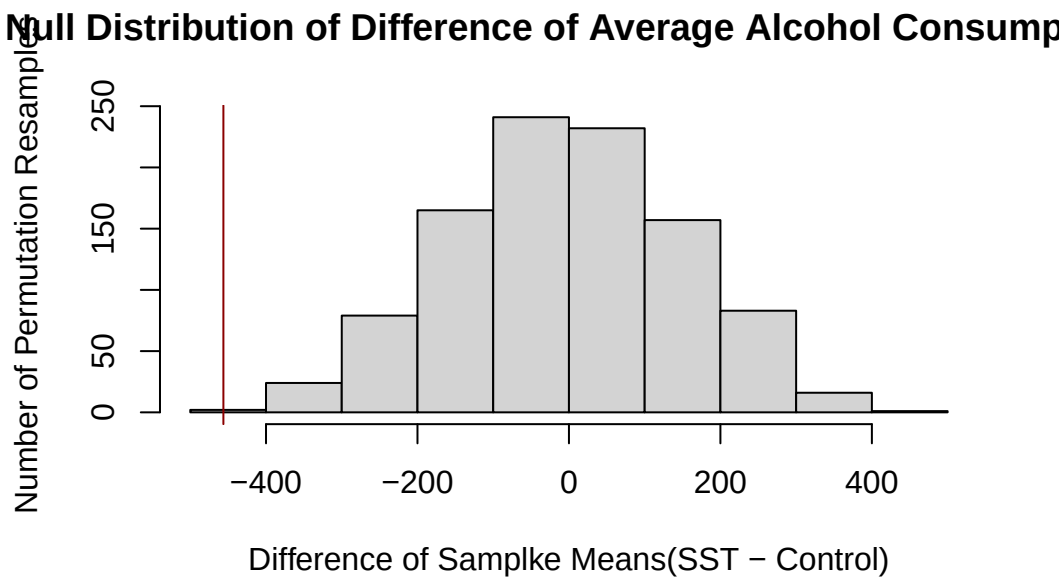
Construct a graph showing the permutation null distribution of t-statistics. Use 999 random permutation resamples. Include a vertical line at the observed value.

```r
names(alcohol)
```

```
[1] "Group"  "Intake"
```

Solution

```r
sim_lab_alc <- read.csv("C:/Users/aocho/Downloads/alcohol.csv")
set.seed(138)
SST_patients <- which(sim_lab_alc$Group == "SST")
xbar_SST <- mean(sim_lab_alc$Intake[SST_patients])
xbar_control <- mean(sim_lab_alc$Intake[-SST_patients])
obs_lab_diff_mean <- xbar_SST - xbar_control

#Find simulated difference means.
B = 999
sim_lab_diff_mean <- numeric(B)
set.seed(138)

for(i in seq_along(sim_lab_diff_mean)){
  resample <- sim_lab_alc
  resample$Intake <- sample(resample$Intake)
  SST_patients <- which(resample$Group == "SST")
  xbar_SST <- mean(resample$Intake[SST_patients])
  xbar_control <- mean(resample$Intake [-SST_patients])
  sim_lab_diff_mean[i] <- xbar_SST - xbar_control

}
#Find permutation null distribution
all_lab_diff_mean <- c(obs_lab_diff_mean, sim_lab_diff_mean)

#Create plot
hist(
  all_lab_diff_mean,
  main = "Null Distribution of Difference of Average Alcohol Consumption",
  xlab = "Difference of Samplke Means(SST - Control)",
  ylab = "Number of Permutation Resamples"

)
abline(v = obs_lab_diff_mean, col= "red4")
```



Part c (Code: 0.5 pts)

Using the permutation null distribution, obtain the p-value for the two-sided test.

Solution

```
# two-sided permutation p-value

p_value <- (sum(abs(sim_lab_diff_mean) >= abs(obs_lab_diff_mean)) + 1) /
           (length(sim_lab_diff_mean) + 1)
p_value
```

```
[1] 0.001
```

Part d (Explanation: 1 pt)

Is there is convincing evidence of a difference in population mean alcohol consumption between patients using the SST program and patients using a traditional treatment program? Why or why not?

Solution

If there were no true difference in population mean alcohol consumption between SST patients and traditional-treatment patients, we would observe a difference in sample means at least this large only about 0.1% of the time due to random chance alone.That is very strong evidence against the null hypothesis.

patients, we would observe a difference in sample means at least this large only about 0.1% of the time due to random chance alone.

That is very strong evidence against the null hypothesis.